MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 8765

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1

Mounting position: Any

Weight: 7 ounces (198.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Ebb	Ecc	egy	egx	epy	epx	Ib
Unit:	Vac	Vdc	Vdc	v	v	kv	kv	Adc
Maximum:	6.8	---	150	600 (see note 3)	200	12.0 (see note 1)	12.0 (see note 2)	0.2
Minimum:	5.8	300	---	175	---	---	5% epy	---
TEST CONDITIONS:	6.3	---	0	150	---	12.0	---	---

ABSOLUTE RATINGS:

Parameter:	ib	Ip	dik/dt	Pb	prr	tj	tk	TA
Unit:	a	Aac	a/µs	---	---	µs	sec	°C
Maximum:	350	5.0	2,000	4.0×10^9	---	0.005	---	+150
Minimum:	---	---	---	---	---	---	180	-55
TEST CONDITIONS:	---	---	---	---	1,500	---	180	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3267	Instantaneous starting	4	epy = 12 kv (min); Ef = 6.8 Vac	0.65	II	---	---	---	---
(A) 3246	Operation (1)	5	Ef = 5.8 Vac; epy = 14 kv; t = 10 minutes	0.65	II	egy	---	150	v
3247	DC anode voltage for conduction	-	Ef = 5.8 Vac	0.65	II	Ebb	---	300	Vdc
3251	Pulse emission	-	ik = 350 a; tp = 5.0 µs ±10%; prr = 60 ±10%; tr = 0.5 µs (max); specified time interval = 2.5 µs	0.65	II	egk	---	200	v
(A) 3241	Heater current	-		0.65	II	If	4.5	9.5	Aac
	Quality conformance inspection, part 2								
(A) 3246	Operation (1A)	-	Operation (1); Ef = 6.8 Vac; tk = 5 minutes; t = 10 minutes;	---	---	egy	---	150	v
3256	Anode delay time	-	Operation (1); t = 120 seconds	---	---	tad	---	0.5	µs
3256	Anode delay time drift	6	Anode delay time	---	---	Δtad	---	0.10	µs
3261	Time jitter	5,7	Operation (1); except epy = 3 kv	---	---	tj	---	0.005	µs
	Quality conformance inspection, part 3								
---	Life test	5	Group C; t = 96 hours "on" and 1 hour "off" (tube mounted in horizontal position); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
3246	Operation (1) and (1A)	-		---	---	egy	---	150	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	300	Vdc
3261	Time jitter	-	egy = 150 v	---	---	tj	---	0.005	µs
1031	Variable-frequency vibration	8,9	No voltages applied	---	---	---	---	---	---
1041	Shock	8	100 G; no voltages applied	---	---	---	---	---	---
---	Shock and variable-frequency vibration end points:								
3246	Operation (1)	-		---	---	---	---	---	---
3261	Time jitter	-		---	---	tj	---	0.005	µs
(A) 3246	Operation at elevated ambient temperature	5,8, 10	TA = 150°C; t = 5 hours	---	---	egy	--		

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NOTES:

1. Instantaneous starting is permissible. The maximum permissible epy is 8 kv and shall not be attained in less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs, maximum duration, shall not exceed 2.0 kv during the first 25 µs following the pulse.

3. The driver pulse, measured at the tube socket with thyratron grid disconnected: 175 volts minimum, 600 volts maximum; tr = 0.35 maximum; grid pulse duration 2.0 µs minimum. Impedance of drive circuit: 50 to 1,500 ohms. At -55°C, a minimum of 250 volts is required.

4. This shall be the first test after the holding period. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 8 kv minimum within 0.03 second. (The filter in the rectifier shall be so designed that the epy reaches at least 4 kv within 0.015 second). The interval between successive attempts to instantaneously start the tube shall be not less than 10 nor more than 30 seconds. Any tube failing to start within three attempts shall be considered a failure.

5. The circuit constants shall be so chosen, that at epy = 14 kv under resonant charging conditions; dik/dt = 1,500 a/µs(min); ib = 150a(min); tp = 1.0 µs ±10 percent; prr = 1,000 (min); Rgl = 30,000 ohms. Grid pulse characteristics shall be: tp = 2.0 µs(max); tr = 0.34 µs(min); and driver impedance = 500 ohms(min).

6. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 and 10 minutes of the operation (1) test. The change in anode delay time (with respect to the 2 minute reading) shall not exceed the specified value.

7. The tube shall be tested by applying a peak forward anode voltage not to exceed the value specified in the test conditions for the time jitter test immediately after the cathode warmup period (tk). After 60 seconds of operation the variation in firing time (tj), shall be not greater than the amount specified.

Ⓐ 8. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

9. There shall be no pronounced resonance in the range from 0 to 2,000 pps.

10. This test shall be conducted for a total of 5 hours with no more than three kickouts and no detrimental anode heating. The tube shall be started with 107.5 percent Ef and operate at this value for 4 hours. At the start of the fifth hour, and while the tube is still operating, the voltage shall be lowered to Ef = 92-1/2 percent and remain there for the final one hour of operation.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 80
DSA - ES

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG
Air Force - 11

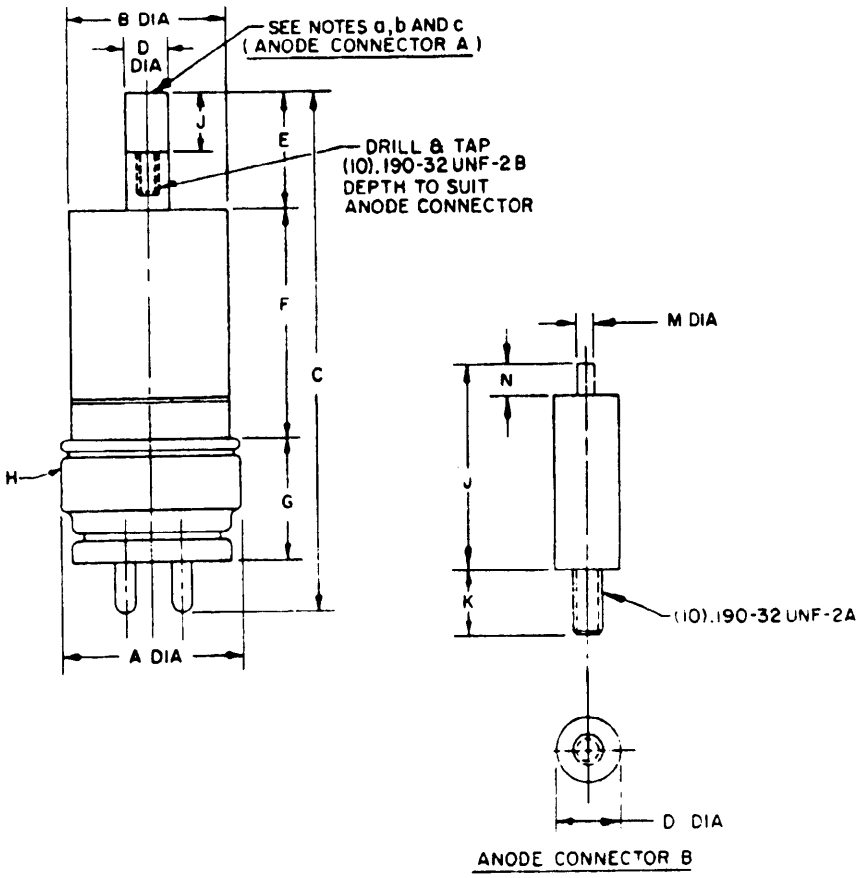
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2989)

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Insoluble matter in toluene



Pin connections	
Pin No.	Element
1	h
2	k
3	g
4	h & k
cap	a

FIGURE 1. Outline drawing of electron tube type 8765.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	--- ---	1.630 (41.40)
B	1.315 (33.40)	1.435 (36.45)
C	4.27 (108.46)	4.77 (121.16)
D	.355 (9.02)	.365 (9.27)
M	.077 (1.96)	.083 (2.11)
N	.188 (4.78)	--- ---
Quality conformance inspection, part 3 (SEE NOTE d)		
H	Base: A4-102 (EIA)	
Reference dimensions		
E	1.00 (25.40)	
F	1.969(50.01)	
G	1.062(26.97)	
J	.50 (12.70)	
K	.375 (9.53)	

NOTES:
a. Types 8765 and 8370, use anode connector A.
b. To replace type 5957, remove anode connector A.
c. To replace type 5956, remove anode connector A and replace with anode connector B.
d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8765 - Continued.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-28254

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